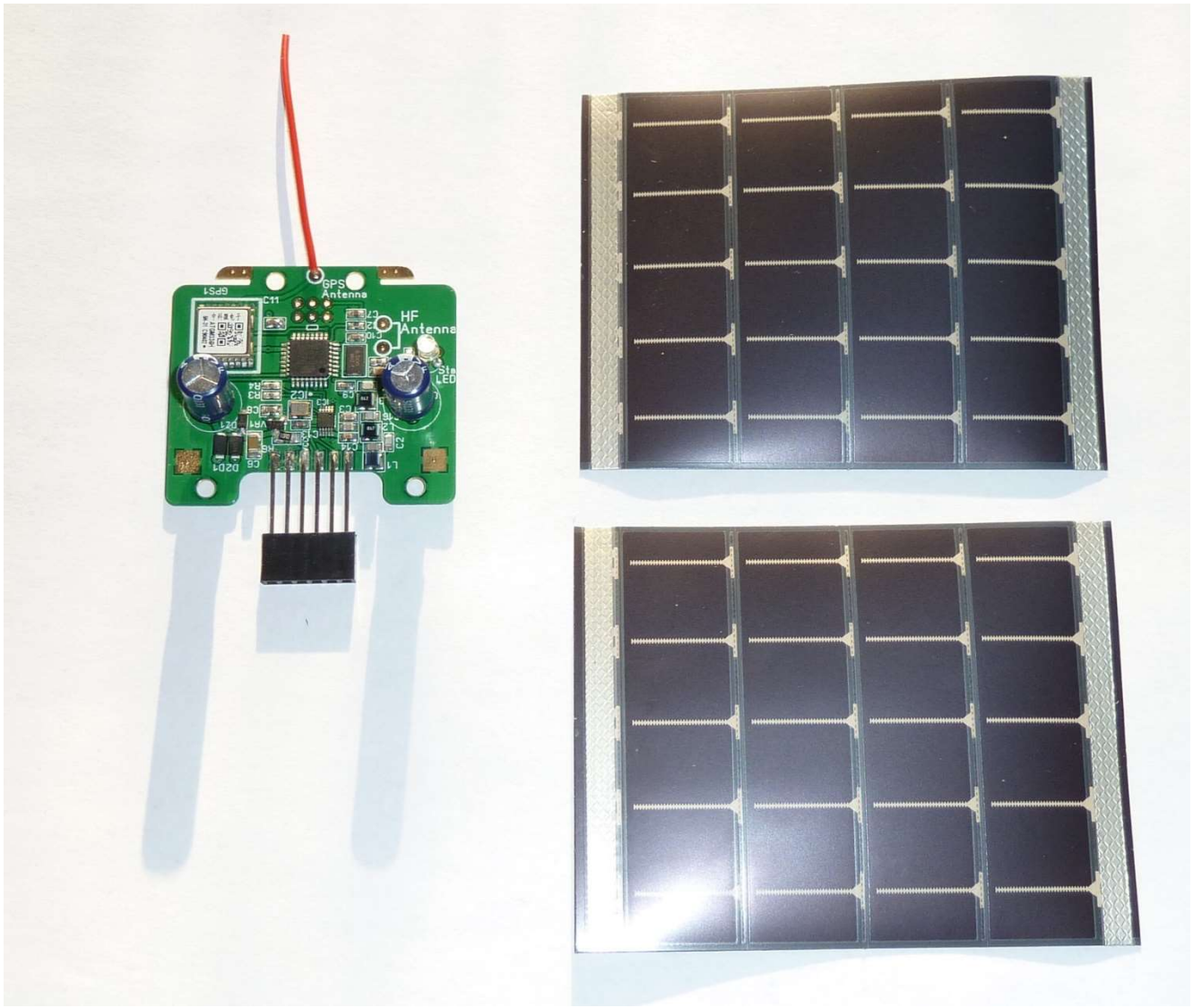
1028 WSPR-TX Pico Assembly Instructions.



The WSPR-TX Pico (product number 1028) is a lightweight WSPR transmitter suitable for small hydrogen or helium balloons.

The WSPR Pico comes built and tested but for shipping reason the solar cells are not attached. This document describes how to solder them to the transmitter and make the transmitter flight ready.

Skills required:

1. Soldering skills.

Tools and materials required:

1. Wire cutter.
2. Soldering iron and solder
3. Copper wire

Optional – Hobby knife (Scalpel)

Estimated time for build:

20 to 60 minutes

Preparation.

Preparing and setting up a work area.

Make sure you have the following items at hand:

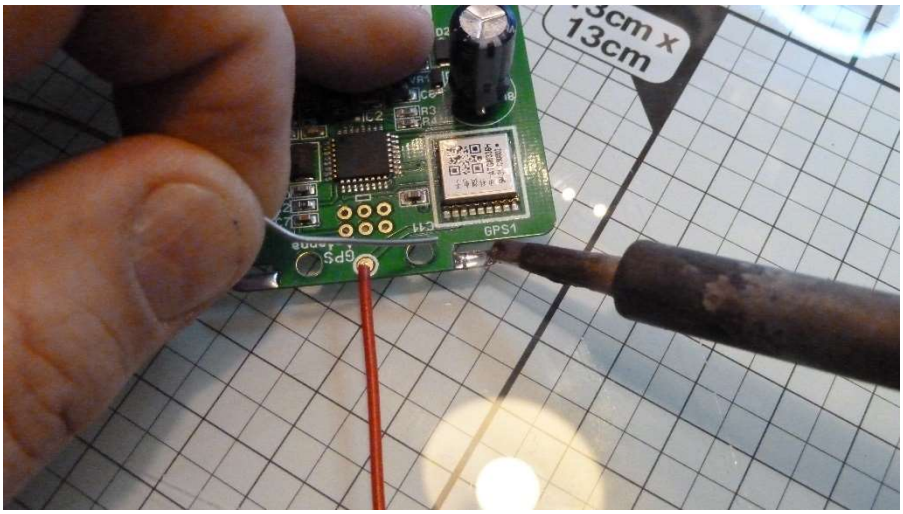
1. The WSPR-TX_Pico board and the two solar cells
2. The tools and materials
3. Workspace

Get a clear table space with a soldering iron at hand. If you print this instruction have it next to you or have a computer next to you if you read it as you go along.

Solar cell assembly.

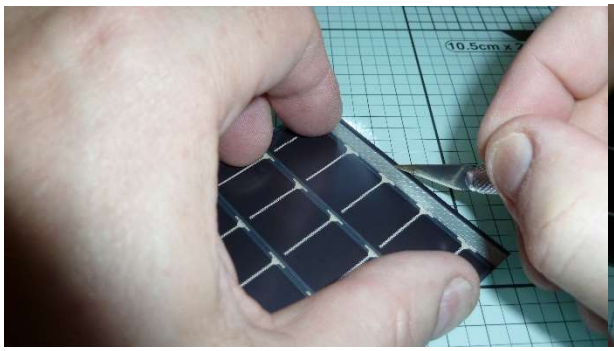
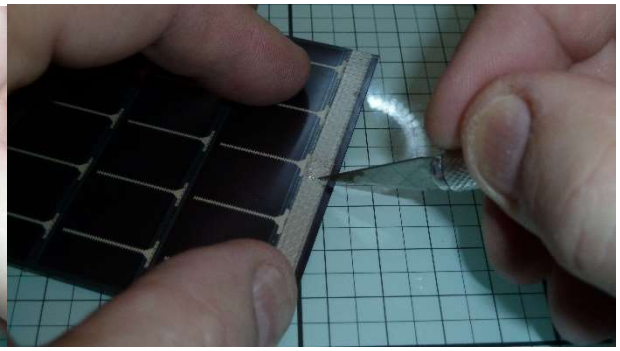
Preparing the surfaces with solder.

1. Add solder to the top tabs on the Pico.
   With a soldering iron heat and add a generous blob of solder on both sides of the tabs on the top of the transmitter as shown in the picture.

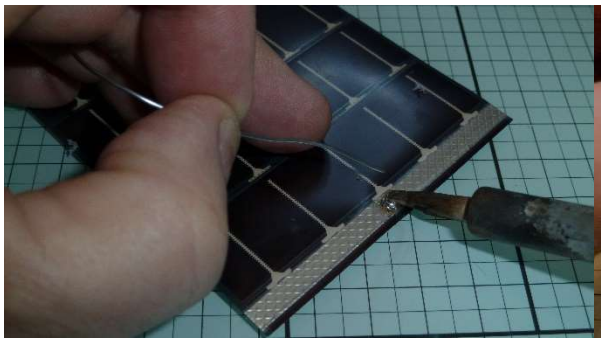
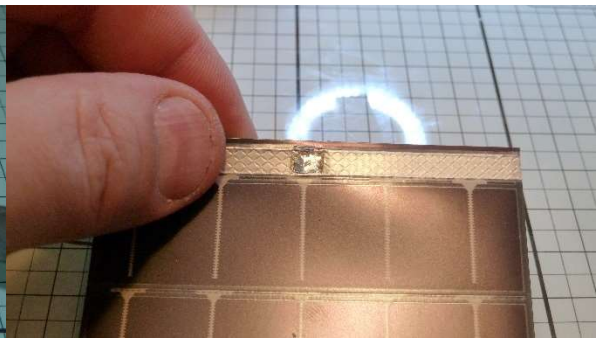


2. Add solder to the mid of the metal strips on the solar cells.

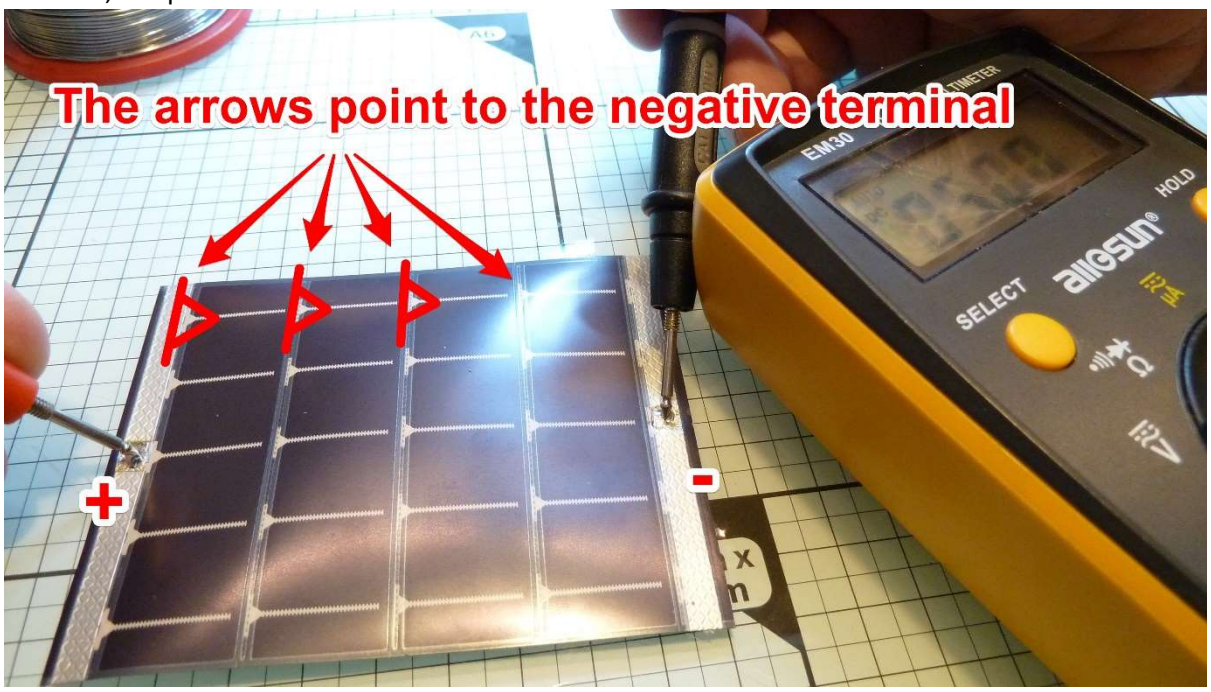
2.1. OPTIONAL STEP The metal strip on the solar cells is coated with plastic. One can burn thru this with a soldering iron and this is the recommended method from the solar cell manufacturer but I like to prepare it by using a scalpel and cut and scrape of some of the plastic first.
This in an optional step and can be skipped.
In the perfect center of the metal strips at both ends of the solar cells remove the plastic with a scalpel. First gently cut a small square in to the plastic, then scrape this square area as shown in the pictures, this will remove most of the plastic.

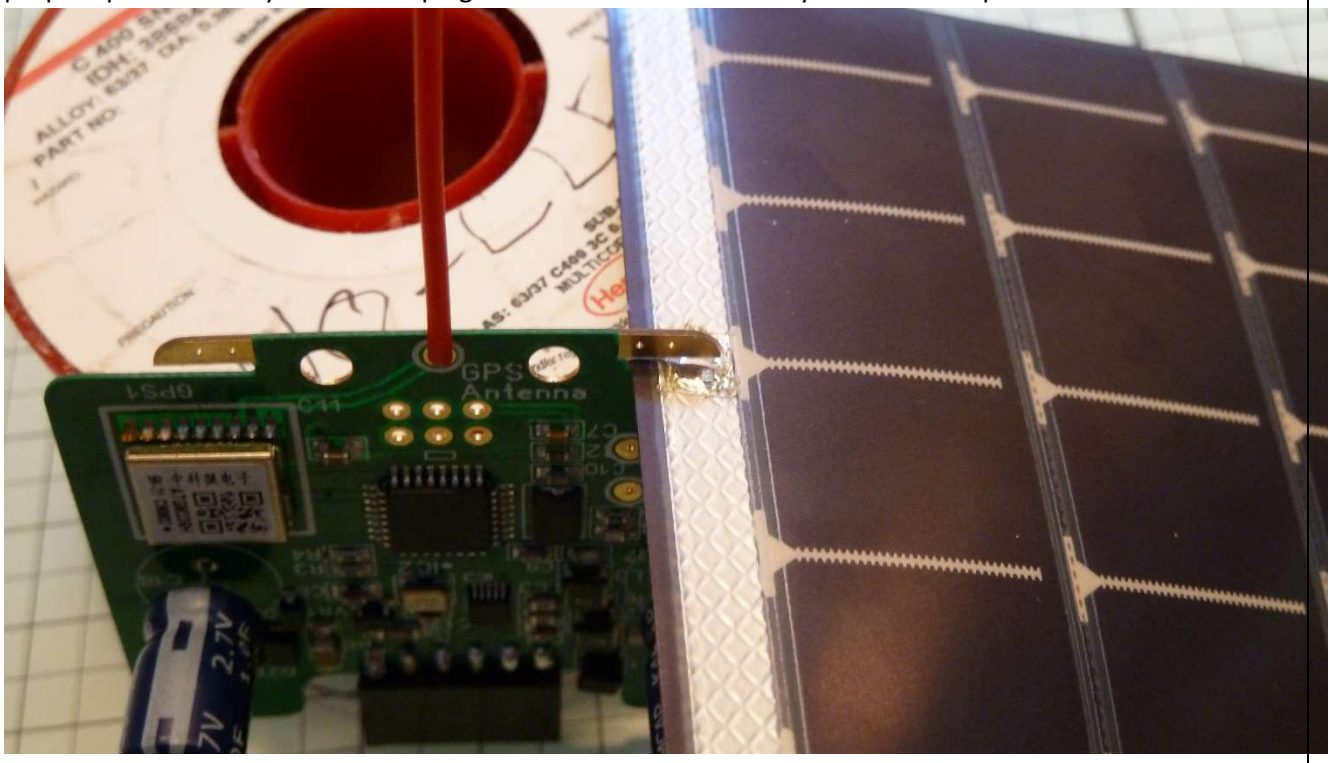
2.2. Take a solder iron to the exact center on both metal strips and heat until you burn thru the plastic. Add some solder.

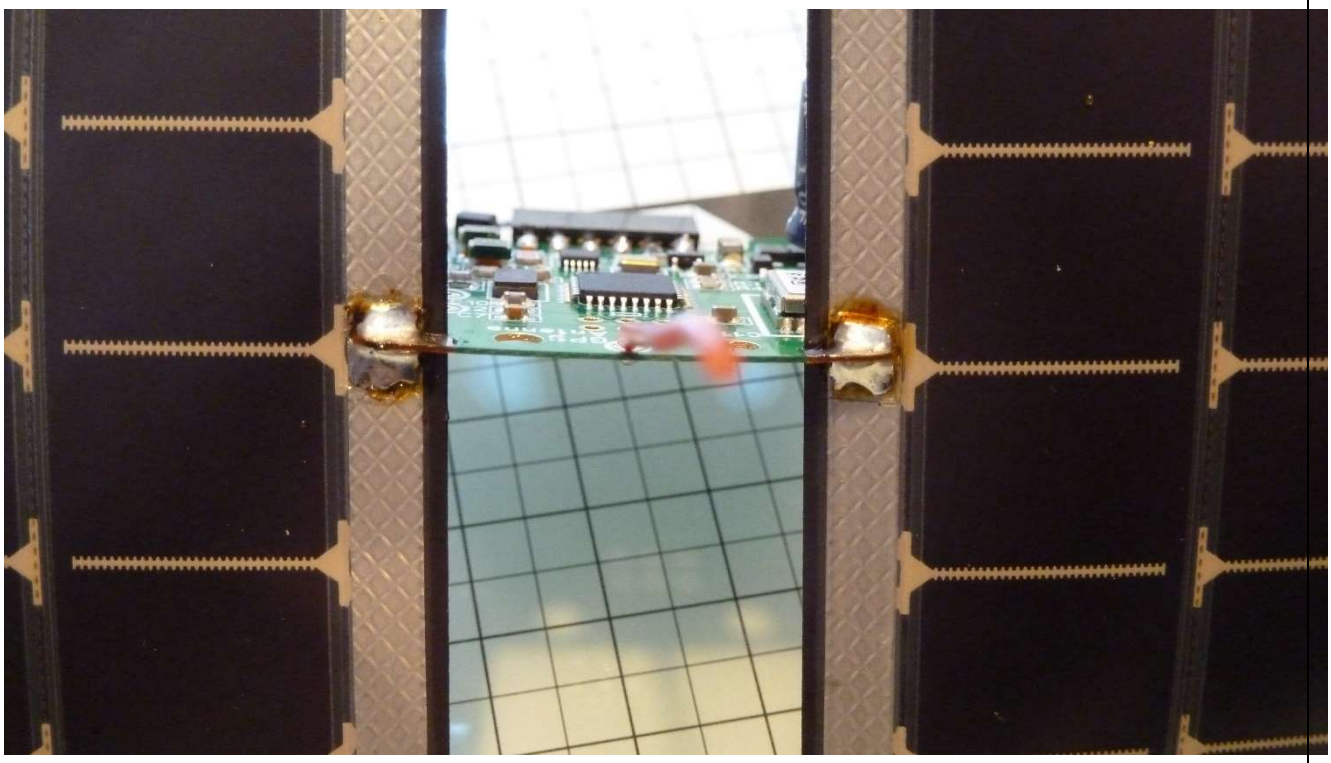
3. Solder the solar cells to the top of the transmitter.
   3.1. Find the positive side of the two metal strips by either using a voltmeter or by looking at the pattern on the solar cells and identify what looks like small arrows. The arrows point towards the negative terminal, see picture.



3.2. Slide the middle of the metal strip on the positive side in in the cut on top the PCB. You may need to prop it up in some way or use a helping hand from someone while you solder it in place.

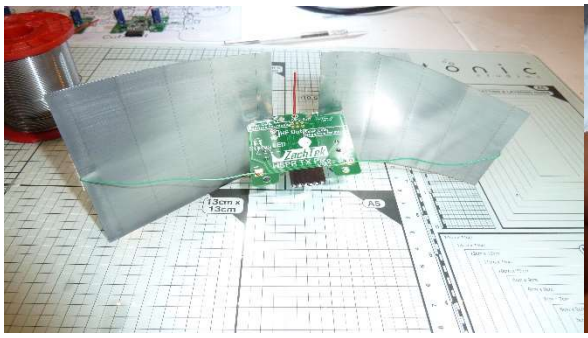


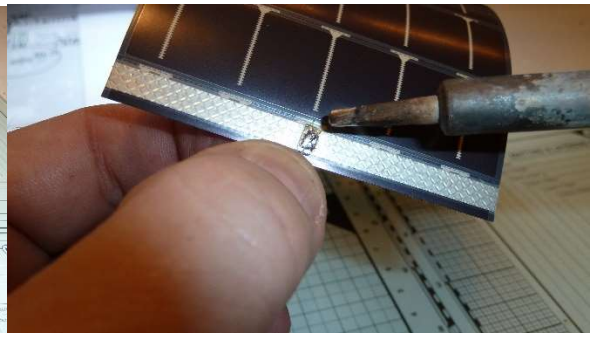
Both solar cells soldered in.



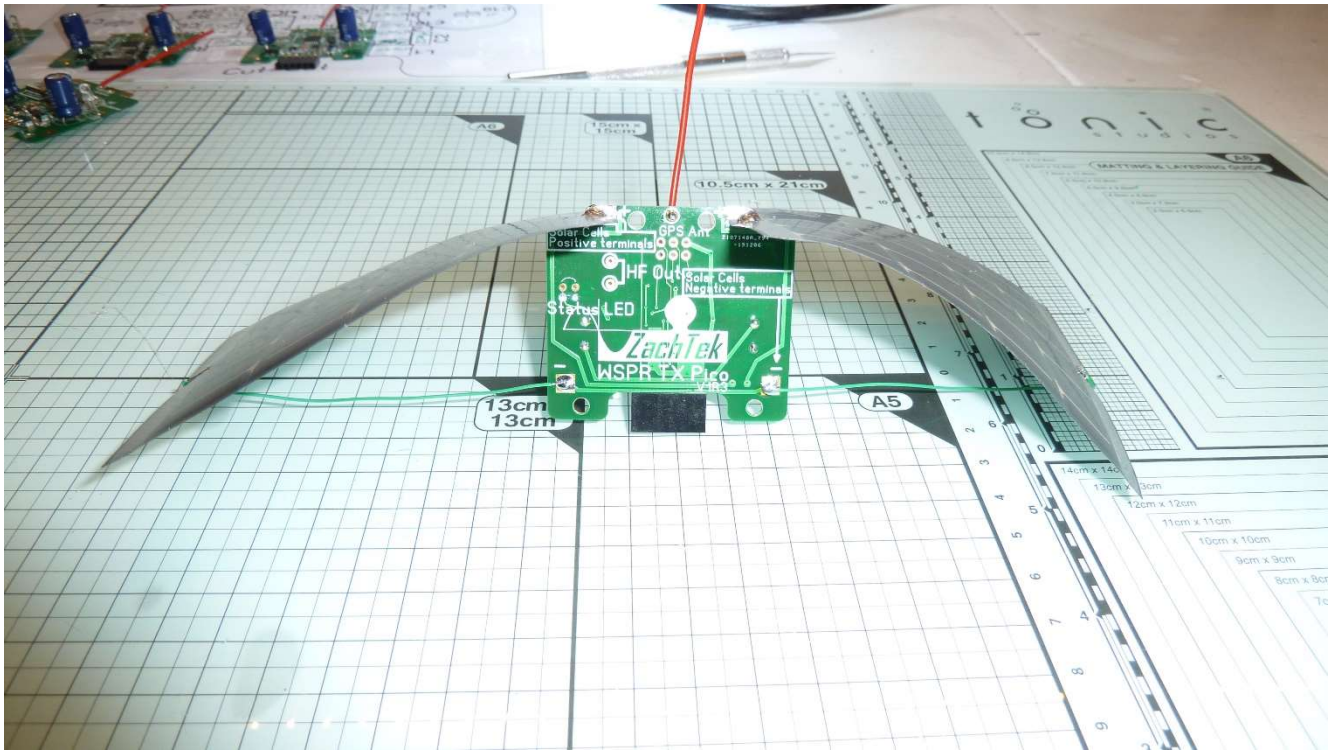
4. Solder a wire to the negative side of the solar cells to the board.
   4.1. Prepare two wires about 9cm long each (3.5Inch), they can be as thin as you have as they only carry max 50mA each.
   First solder them to the golden squares at the bottom part of the transmitter and then trim the wire to length so the solar cells slope downward a bit when they are stretched to the solar cells

4.2. Solder the other ends of the wires to the bottom of the solar cells.



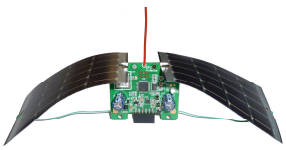


The slope helps the transmitter to operate when the sun is lower in the sky. Depending on latitude and time of year, you might prefer more or less slope. It is not super critical and I would recommend not overdoing it and keep them just slightly tilted downward.
If you happen to get one solar cell sloping a bit more than the other like in the picture below then don't worry about it, the transmitter will most likely spin constantly so the solar angle will be dived between the two cells and at different time of the day the sun will favor one or the other.



As the transmitter spins the cell might be in temporary shadow, the Super Capacitor will then use its stored energy to power the transmitter.
When the two super caps are fully charged they alone can power the transmitter for up to 30 seconds.
One solar cell is theoretically enough to power the transmitter- having two ensures that the power budget stays positive for more varying solar angles.
10.5 hours of operation have been observed on a summer day over Australia at 11km altitude.

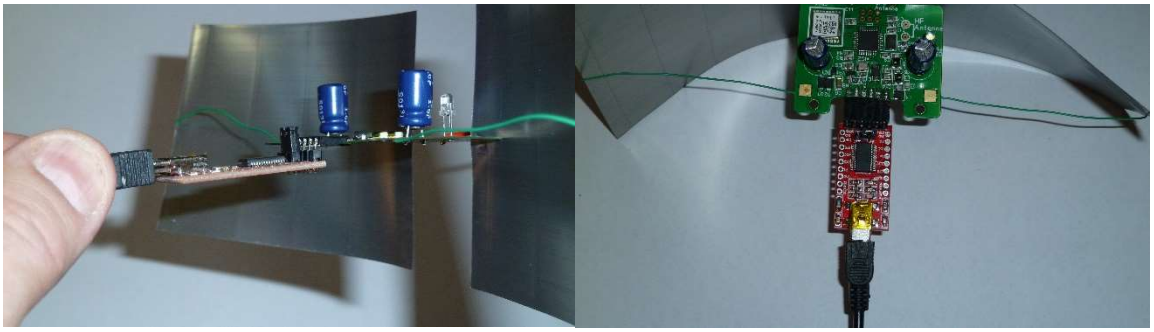
OK you have now made it possible to power the transmitter from sunlight!
Let us test it out before you do the harness and HF Antenna.



Testing the operation.

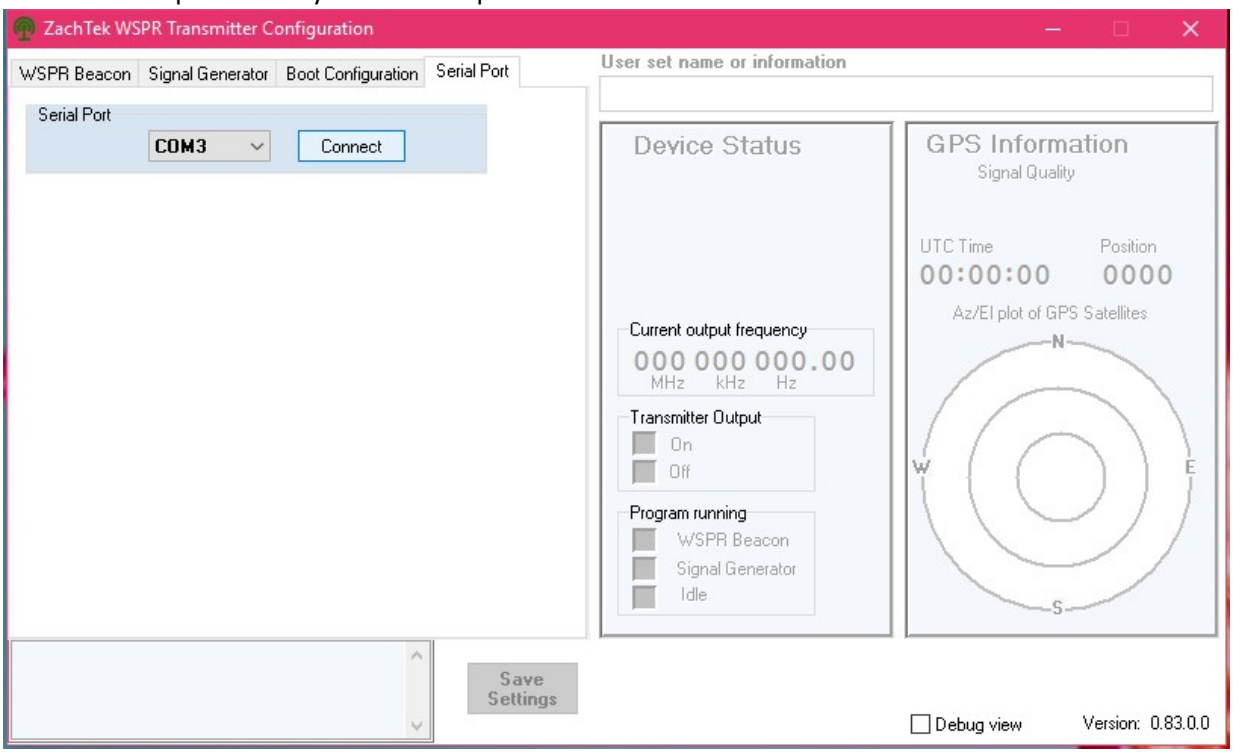
Making sure it is configured with a call sign before testing it in sunlight.

1. The transmitter needs to have a call sign configured or else it will not transmit so if you have not configured it before it is time to do it now.
   Bring out the USB to serial converter and plug it in to a computer at one end and to the transmitter on the other end as shown in the pictures.
   This converter will power the Transmitter when it is connected to a computer and will enable configuration with the Windows software

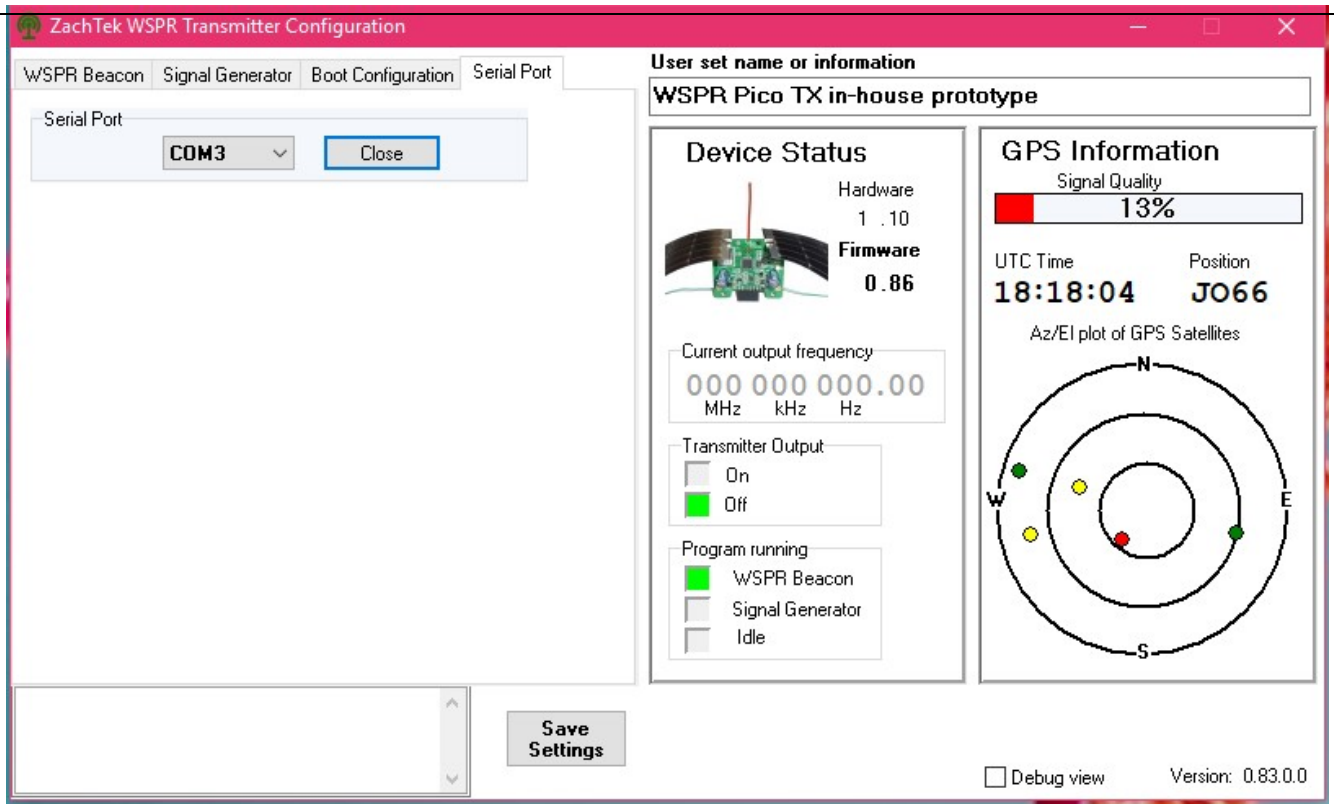


   The USB to serial adapter has a jumper that sets either 5V or 3.3V output, the transmitter can operate in either setting but it is recommended to have it in the 5V setting.

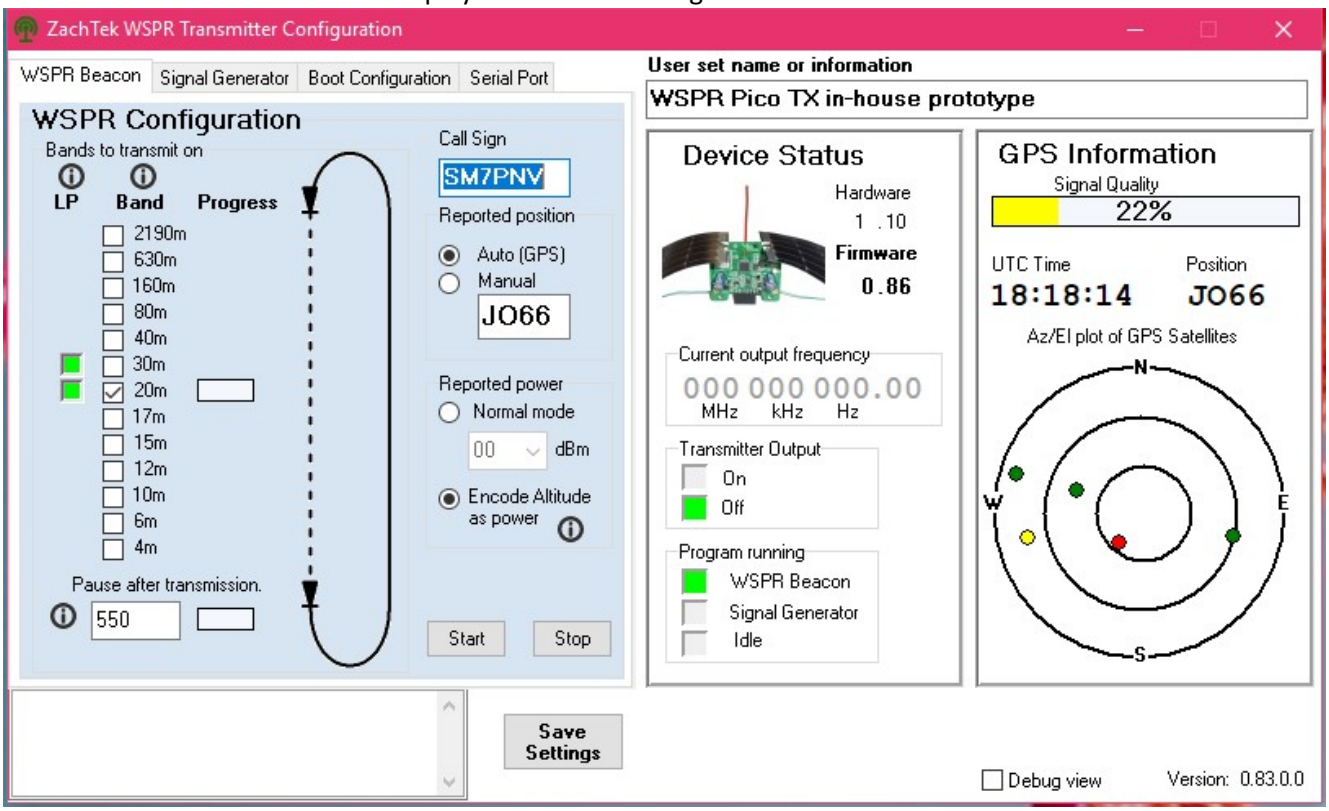
2. Download the PC Configuration program from zachtek.com/download and start it.
3. Pick the serial port used by the USB adapter and click the “Connect” button



   If the correct port was used you will see Device status information that shows it is the WSPR Pico you are communicating with. If not try another port until you find the correct one.



4. Click the “WSPR Beacon” Tab to display the Beacon settings.



Fill in you call sign and pick what bands to transmit on. Transmitting on both the 20m and 30m bands is recommended even if a simple dipole antenna will favor just one of the bands (witch one depends on the length of the antenna)

Also, set a suitable transmission paus. 400 seconds is a good setting that allows transmission every 10 minutes.

5. Click the save settings button.

6. Optionally run the signal generator or the WSPR beacon to try out the transmitter.
   Even with no HF antenna attached, it should be possible to pick up the signal on a nearby receiver.
   The GPS with its simple quarter wave antenna can have a hard time operate indoors so it might take a very long time before it get a fix and start the WSPR transmission. You can watch the GPS info in the right portion of the screen but it might be better to take it outdoors to check the operation as the GPS reception will be so much better outdoors.

Testing outdoors.

1. Take the transmitter outdoors on a sunny day. You should have no computer and no USB to Serial adapter attached, just hold the transmitter in your hand and point the solar cells to the sun while watching the White status LED.

2. After about a minute, the White Status LED should start flashing.
   The flashes has the following meaning:
   16 Rapid flashes = Booting up
   1 flash every second = Running WSPR Beacon – waiting for GPS fix
   2 flashes every second = Running WSPR Beacon, fix acquired – waiting for top of even minute.
   6 rapid flashes every 0.6 seconds = Running WSPR Beacon - actively transmitting
   1 flash every 9 seconds =Running WSPR Beacon, having a transmission pause - Microcontroller and GPS is sleeping to save power

Frequently Asked Questions:

Q: I would like to have a temporary antenna connector so I can measure the output or connect to a ground-based antenna, can I do that?
A: Yes, the “HF Out” output pads is a 50ohms port so just solder in your connector to it with a couple wires. The solder pad closest to R5 is the hot output and the other is ground.
See picture for an example of a SMA connector soldered in to the output.

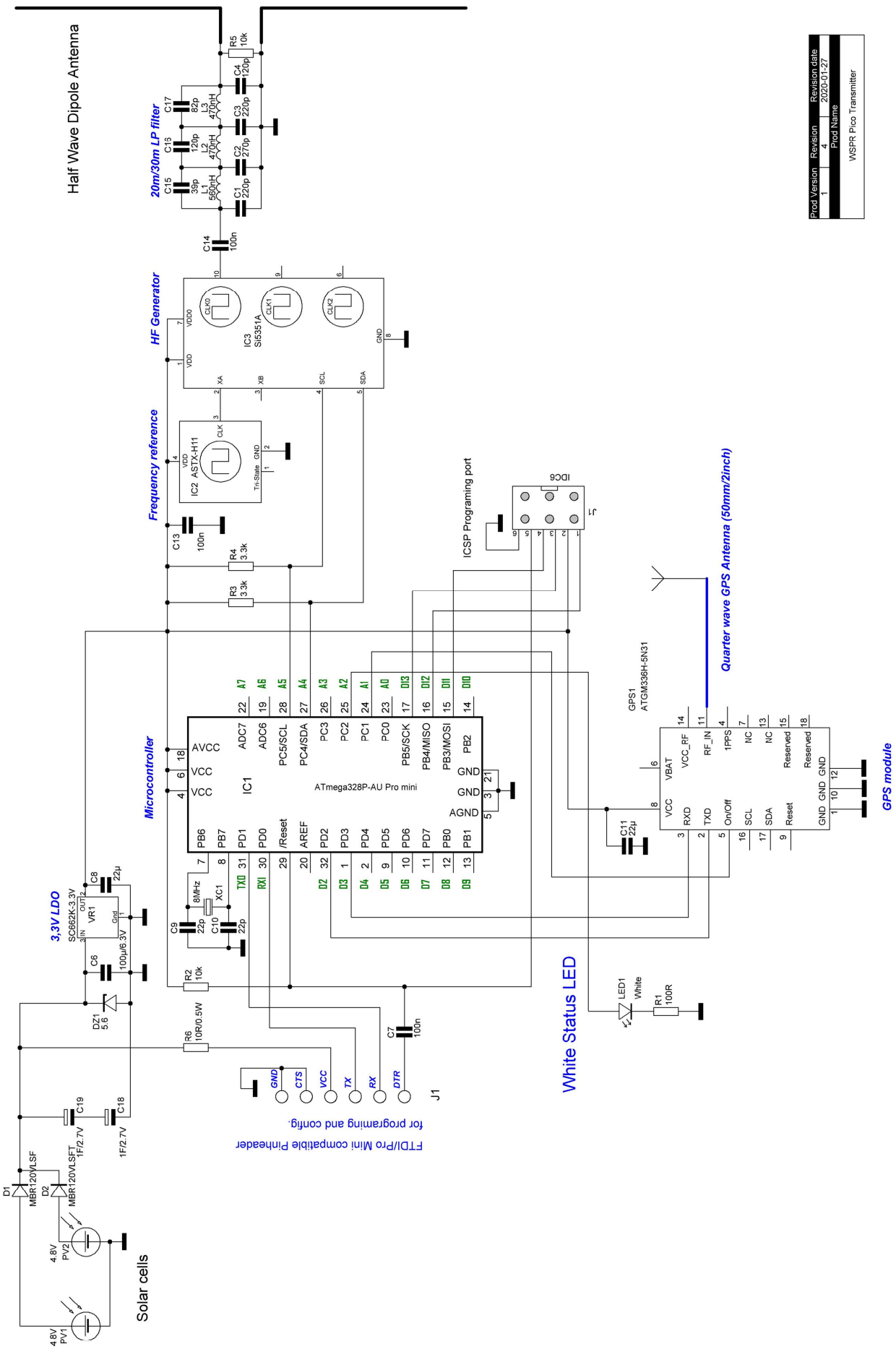
Q: I would like to use it on another band than 20m or 30m can I do that?
A: Yes but you might have to bypass the low pass filter.
If the band you want to operate on is lower in frequency (like for example 7MHz the 40m band) you don’t have to do anything with the existing low pass filter, just leave it in place and provide your own low pass filter externally to suppress the overtones.
If you want to operate above 14MHz you need to remove L1,C1 and C15 to disable the low pass. You can then solder in your output wire to the pad of L1 or to C14

Appendix.

Altitude to Power table:

Power in dBm	Power in Watt	Altitude meter	Altitude feet
0	**1mW**	**<900m**	
3	**2mW**	**0.9-2.1km**	
7	**5mW**	**2.1-3km**	
10	**10mW**		
13	**20mW**		
17	**50mW**		
20	**100mW**		
23	**200mW**		
27	**500mW**		
30	**1W**		
33	**2W**		
37	**5W**		
40	**10W**		
43	**20W**		
47	**50W**		
50	**100W**		
53	**200W**		
57	**500W**		
60	**1kW**	**>18km**	

Schema.


Half Wave Dipole Antenna
20m/30m LP filter
HF Generator
Frequency reference
Microcontroller
ATmega328P-AU Pro mini
3,3V LDO
ICSP Programing port
Quarter wave GPS Antenna (50mm/2inch)
GPS module
White Status LED
Solar cells
FTDI/Pro Mini compatible Pinheader
for programing and config.
Prod Version
Revision
Revision date
Prod Name
WSPR Pico Transmitter

END OF DOCUMENT